

Mohammed Ashfaq

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SUMMARY

AI/ML Engineer with 2+ years of hands-on experience building production-grade generative AI solutions, including LLM-based chatbots and agentic validation pipelines. Proven ability to design, optimize and deploy deep learning models on AWS, implement prompt-engineering strategies, and deliver end-to-end AI workflows using Python and modern ML frameworks. Strong collaborator who translates business problems into data-driven AI products while maintaining rigorous data quality and documentation standards.

TECHNICAL SKILLS

Data & Analytics: Data Structures & Algorithms, Statistical Modeling, Data Visualization (Matplotlib, Seaborn, Plotly)

Frameworks & Tools: TensorFlow (conceptual experience via vLLM), PyTorch (conceptual experience via Llama 3.1), Keras (conceptual experience), Temporal, Docker, Podman, AWS, Prometheus, Grafana, DCGM Exporter, cAdvisor, Agile/Scrum

Machine Learning & Deep Learning: Generative AI (Llama 3.1, GPT-style models), vLLM, Hugging Face, RAG, Prompt Engineering, NLP

Programming Languages: Python, Elixir, Javascript, HTML, CSS

EXPERIENCE

TechnoIdentity

AI/ML Engineer, Hyderabad

Jun 2024 - Present

- Architected and delivered a generative AI platform for engineering-drawing validation, leveraging Llama 3.1 (GPT-style) and Retrieval-Augmented Generation to automate domain-specific insight extraction.
- Implemented prompt-engineering pipelines and confidence-based feedback loops that auto-re-run validation when confidence scores fell below 90%, improving model reliability and business decision accuracy.
- Utilized Temporal durable workflows to orchestrate long-running LLM pipelines, providing fault-tolerant, production-ready AI microservices on AWS and reducing inference costs by ~40%.
- Developed an Elixir SDK with Rust bindings to expose Temporal's workflow engine, enabling scalable microservice orchestration and reinforcing software engineering best practices (Agile, CI/CD).
- Built a privacy-first, on-premise AI chatbot powered by Llama 3.1 8B and Hugging Face RAG techniques, demonstrating deep-learning model deployment, NLP processing, and multi-modal data handling.
- Optimized model performance through load testing, concurrency tuning, and GPU utilization monitoring with Prometheus, Grafana, DCGM Exporter, and cAdvisor, ensuring scalability for high-traffic production workloads.
- Led migration from Docker to Podman, strengthening deployment security, compliance, and operational efficiency in cloud-native environments.
- Authored comprehensive technical documentation, user guides, and presentation decks for cross-functional stakeholders, translating complex AI concepts into actionable business insights.

PROJECTS

Agentic AI Engineering Drawing Validator | *Python, Temporal, Llama 3.1, vLLM, AWS, Prometheus, Grafana*

- Designed a generative AI validation engine using GPT-style LLMs and Temporal durable workflows to orchestrate resilient, long-running inference pipelines.
- Implemented prompt-engineering and confidence-based refinement loops that auto-re-execute validation, achieving a 40% reduction in inference cost and higher accuracy.
- Deployed the solution on AWS with production-grade monitoring and auto-scaling, meeting enterprise reliability and latency requirements.

Elixir SDK for Temporal | *Elixir, Rust, Temporal, Docker, Agile*

- Created a full-stack SDK that exposes Temporal's workflow orchestration to Elixir applications, enabling durable microservices and long-running AI business workflows.
- Integrated retry, timeout, and task-scheduling primitives, facilitating robust AI pipeline execution in production environments.

E-Learning Platform AI Chatbot | *Llama 3.1, Hugging Face, Retrieval-Augmented Generation, AWS, Podman, Prometheus,*

- Built a privacy-first, on-premise chatbot using a GPT-style model and RAG, delivering accurate domain-specific answers for e-learning content.
- Engineered end-to-end CI/CD pipelines, performance testing, and observability stack to ensure high availability and compliance with enterprise data policies.

EDUCATION

Osmania University

B.E. Electronics and Communications, Hyderabad, Telangana, India

2018 - 2022

- GPA 7.12/10